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Mining—The dark side of the energy transition

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ABSTRACT

The energy transition will significantly increase demand for minerals. This might create new economic opportunities for lower-income countries with resources, but it will also augment local social and environmental problems associated with the sector. The increasing number of social and environmental conflicts associated with mining in emerging economies is a manifestation of this tension and raises questions about just energy transitions. But can mining be done in more sustainable ways? Transition studies have been largely silent on this question. In this article, we urge transitions scholars to incorporate mining into their analyses of energy transitions and briefly mention possible starting points to move in this direction.

The energy transition seems to be finally happening. Excellent news. Nevertheless, this transition has a dark side. Different estimates indicate that clean energy technologies require more minerals. Wind plants consume nine times more minerals than gas-fired power plants and electric cars use six times more minerals than conventional cars (IEA, 2021; Hund et al., 2020). Concerns are being expressed about the global environmental impact of this shift (Klare, 2021) and the multiple challenges for the feasibility of energy transition that the increased dependency on minerals will bring (IEA, 2021; World Bank, 2017). Nevertheless, far less attention has been paid to the impact that the transition will have on emerging economies that depend on these minerals, and more importantly, to how they should deal with the development tensions that will be generated by the increasing demand.

Mining is a central economic activity for many emerging countries (e.g. Mongolia, Guinea, Botswana, the Democratic Republic of the Congo, Chile, Peru and South Africa). For these countries, the increase in the demand for minerals will involve the opportunity for higher investments, exports, employment and fiscal revenues. Nevertheless, the multiple local environmental and social problems that mining activities generate will increase too: local pollution of soil, air, and water; the disposal of toxic residuals; intensive use of water and energy (IEA, 2021), work and environmental risks, child labour and sexual abuse (Sovacool, 2021), and corruption and armed conflict (Church and Crawford, 2018).

If concerns about "just transitions" are real, therefore, we need to take seriously the question of how to support transitions in the whole energy system, which is intrinsically linked to the material-supplying system of which mining is a central part (see Raven and Verbong, 2007; Schot and Kanger, 2018). Discussions about jobs lost in phasing-out industries or about how to dematerialise growth in advanced economies are important, but they do not address "the difficult questions surrounding the sourcing and supply of minerals and metals required to support renewable energy-systems in a low-carbon economy" (Bainton et al., 2021).

From the perspective of mining economies, some of these difficult questions about transitions are: How to support and sustain transitions in more sustainable directions in the mining sector? Is it possible to support the deployment and development of mining

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technologies and organisational forms that respect the environment and local livelihoods, and if so how? Which non-technological dimensions of the current mining regime could open up spaces for paths of change?

Emerging economies that specialise in mining do not have the option in the short, or even the medium term, to shut these sectors down. The world economy will at the same time find it impossible to replace the minerals required for the energy transition overnight. These questions, therefore, are central to debates on energy transitions if those transitions are not going to be just about shifting environmental problems towards less advanced countries (van den Bergh et al., 2015).

Innovation studies of development have investigated processes of structural change away from problematic natural resource industries, like mining, but focusing mostly on market and knowledge-related forces for change and stability (Andersen and Wicken, 2020; Marín and van Zwanenberg, 2021). Transition studies could add important insights to this literature by incorporating a better understanding of non-market, and non-cognitive forces for change and stability, for example, political processes and agency (Rip and Kemp, 1998; Geels, 2014).

Transitions research has, however, been silent on the problems faced by mining economies.

Two main reasons explain this lacuna in our view. First, a narrow geographical and thematic focus. As argued by Kanger (2020), research has focused on a small number of socio-technical systems, and most studies have suffered from "methodological nationalism" (Fuenfschilling and Binz, 2018), although the challenge of understanding mining, especially in its relationship with energy, requires a global approach. Second, as in all research fields, transition studies suffer a form of positive results bias and have focused mostly on "positive trends" (Antal et al., 2020). Sustainability niches in energy or food are evident, but less so in mining, where alternatives are limited to specific technological fixes.

So where do we start?

Traditional demand-side pressures are unlikely to work in a sector that produces materials embodied in other final products that are more visible for final consumers, and whose most problematic activities are distant from those consumers, physically and emotionally. Ideas of "impact investment" or of global green standards are being discussed, but for these not to become mere tools for "greenwashing" (The Economist, 2021) their potential for impact needs to be critically analysed in a sector that is not well monitored by final consumers. There are, however, also other forces pushing for change that need to be better understood. One are the social movements (Smith, 2012) that have been questioning mining for decades. Social conflicts associated with mining have been growing significantly and they are having an important effect in limiting mining activity. In Argentina, for instance, due to social conflicts, mining is forbidden or limited in 7 out of 15 of the provinces with large mineral deposits. But how to use the transformative potential of these local conflicts to move things in different directions also at a global scale? National governments, responding to the demands of civil society, can introduce regulations to limit or control mining activities, but these will not in themselves be sufficient, as companies can simply move to another country. To a large extent, therefore, the transformative potential of local conflicts in a sector dominated by a reduced number of global companies that dominate final and technology markets will depend on the capacity of the system to find innovative ways to link local with global, and market and non-market, forces for change.

It is important and urgent that global attention and commitment is devoted to this challenge. Transition studies could and should play a key role in making that challenge more visible and advancing our understanding about how we might address it.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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